

How Organ Systems Work Together

Every organism, whether a plant or an animal, depends on the coordinated action of multiple organ systems. No single organ or system can sustain life alone. A plant must absorb water through its roots, transport it through its stem, and release it through its leaves — all working together. A human must breathe, circulate blood, move muscles, regulate temperature, and fight infection — all through systems that depend on one another. When one system fails, the effects spread throughout the organism. Understanding how organ systems interact helps explain normal body function, the consequences of injury and disease, and how the body defends itself against infection.

This reading material explains how various plant organs interact to facilitate the transport of materials throughout the plant system. It describes how a damaged part in a system affects the organism using the cases of a torn ACL (skeletal and muscular systems) and an asthma attack (respiratory and circulatory systems). It explains how the nervous and endocrine systems work together to regulate body temperature in response to stimuli. Finally, it explains how vaccines work by relating to the functions of the immune system and lymphatic system in protecting an organism against infectious diseases.

Part 1: Plant Organ Systems – Transport of Materials

The Plant as a System

A plant is a living system composed of organs that work together. The main plant organs involved in transport are the **roots**, **stem**, and **leaves**. Each has a specific role, and each depends on the others.

Plant Organ	Primary Role in Transport
Roots	Absorb water and minerals from soil

Plant Organ	Primary Role in Transport
Stem	Contains vascular tissues that transport materials between roots and leaves
Leaves	Produce glucose through photosynthesis; regulate water loss through stomata

The Vascular Tissues: Xylem and Phloem

Inside the stem, roots, and leaves are two types of vascular tissue that act as the plant's transport highways.

Tissue	What It Transports	Direction	Cell Type
Xylem	Water and dissolved minerals	Upward (roots to leaves)	Dead, hollow cells forming continuous tubes
Phloem	Glucose and other sugars	Both directions (source to sink)	Living cells with sieve plates

How Water and Minerals Move Upward

Process	What Happens	Where It Occurs
Absorption	Root hair cells take in water and minerals from soil	Roots
Root pressure	Water entering roots pushes water upward slightly	Roots and lower stem

Process	What Happens	Where It Occurs
Capillary action	Water molecules cling to xylem walls and to each other	Xylem vessels
Cohesion-tension	Water evaporating from leaves pulls a continuous column of water upward	Xylem from leaf to root
Transpiration	Water vapor exits through stomata, creating the pull	Leaves

Key insight: Transpiration is the main driving force. As water evaporates from leaf cells, it creates tension that pulls the entire water column upward through the xylem. This is called the **cohesion-tension theory**.

How Glucose Moves Through the Plant

Step	What Happens
1. Source	Leaf cells produce glucose through photosynthesis
2. Loading	Glucose is loaded into phloem tubes
3. Pressure flow	Water enters phloem, creating pressure that pushes sap toward sinks
4. Transport	Phloem carries sugars to roots, fruits, seeds, or growing tips
5. Unloading	Sugar is unloaded at the sink for use or storage

Source-to-sink concept: Sugar moves from where it is made (source, usually leaves) to where it is needed or stored (sink, such as roots, fruits, or growing shoots).

How the Organs Work Together: A Summary

Organ	Action	Dependency
Roots	Absorb water and minerals	Depend on leaves for glucose to power active transport
Stem	Transport water, minerals, and glucose	Depends on roots for water and leaves for sugar
Leaves	Produce glucose; release water vapor	Depend on roots for water and stem for transport
Stomata	Regulate gas exchange and water loss	Depend on guard cells and environmental conditions

If one organ fails: If roots are damaged, water cannot reach leaves, photosynthesis slows, and the plant wilts. If leaves are removed, glucose production stops, and roots and stems lose their energy supply.

Part 2: Human Body Systems Working Together

Overview of Major Human Body Systems

The human body contains eleven major organ systems. Each performs a specific function, but all depend on one another.

System	Major Organs	Primary Function
Skeletal	Bones, cartilage, ligaments	Support, protection, movement
Muscular	Skeletal muscles, smooth muscles, cardiac muscle	Movement, posture, heat production
Circulatory	Heart, blood vessels, blood	Transport oxygen, nutrients, wastes
Respiratory	Lungs, trachea, bronchi	Gas exchange (oxygen in, carbon dioxide out)
Digestive	Mouth, stomach, intestines	Break down food, absorb nutrients
Nervous	Brain, spinal cord, nerves	Detect stimuli, coordinate responses
Endocrine	Hypothalamus, pituitary, thyroid, adrenal glands	Regulate metabolism, growth, stress response
Immune	White blood cells, antibodies	Defend against pathogens
Lymphatic	Lymph nodes, lymph vessels, spleen, thymus	Support immunity; filter pathogens and waste
Excretory	Kidneys, bladder	Remove wastes from blood
Integumentary	Skin, hair, nails	Protect body; regulate temperature

How Systems Interact

Interaction	Example
Skeletal + Muscular	Muscles pull on bones to produce movement
Respiratory + Circulatory	Lungs oxygenate blood; heart pumps it to tissues
Digestive + Circulatory	Nutrients absorbed in intestines enter blood and are distributed
Nervous + Muscular	Nerves signal muscles to contract
Nervous + Endocrine	Hypothalamus coordinates rapid nerve responses and slower hormone release
Immune + Lymphatic	Lymph nodes filter pathogens and house lymphocytes
Circulatory + Excretory	Kidneys filter wastes from blood

Key insight: Every system depends on at least one other system. No system works in isolation.

Part 3: How a Damaged Part Affects the Organism

Case 1: Torn ACL (Skeletal and Muscular Systems)

What Is the ACL?

The **anterior cruciate ligament (ACL)** is a band of connective tissue in the knee that connects the femur (thigh bone) to the tibia (shin bone). It stabilizes the knee joint and prevents the tibia from sliding too far forward.

Structures Involved

Structure	System	Function
Femur	Skeletal	Thigh bone; forms upper part of knee joint
Tibia	Skeletal	Shin bone; forms lower part of knee joint
ACL	Skeletal (connective tissue)	Stabilizes the knee; prevents forward sliding of tibia
Quadriceps and hamstrings	Muscular	Contract to extend or flex the knee
Tendons	Skeletal (connective tissue)	Attach muscles to bones

What Happens When the ACL Is Torn

Effect	Explanation
Knee instability	Without the ACL, the tibia can slide forward, making the knee feel like it will "give way"
Impaired movement	Muscles cannot effectively move the joint because the bones are not properly stabilized
Pain and swelling	Damage to tissues triggers inflammation
Reduced athletic performance	Cutting, pivoting, and jumping movements become difficult or impossible
Compensatory strain	Other ligaments, muscles, and the opposite leg may take on extra stress

How This Affects the Whole Organism

Level	Effect
Joint	Loss of stability and range of motion
Muscular system	Quadriceps weaken from disuse; hamstrings may overcompensate
Nervous system	Pain signals sent to brain; proprioception (sense of joint position) impaired
Daily life	Difficulty walking, climbing stairs, or participating in sports

Level	Effect
Long-term	Risk of osteoarthritis if untreated

Key insight: A damaged part in one system (skeletal) disrupts the function of another (muscular), affecting the entire organism's ability to move.

Case 2: Asthma Attack (Respiratory and Circulatory Systems)

What Is Asthma?

Asthma is a chronic condition in which the airways become inflamed and narrowed. During an asthma attack, the muscles around the bronchioles tighten, the airway lining swells, and excess mucus is produced.

Structures Involved

Structure	System	Function
Bronchioles	Respiratory	Carry air to and from alveoli
Alveoli	Respiratory	Site of gas exchange (oxygen in, carbon dioxide out)
Smooth muscle around airways	Muscular	Can tighten or relax to change airway diameter
Heart	Circulatory	Pumps blood to lungs and body
Blood vessels	Circulatory	Carry oxygen-poor blood to lungs and oxygen-rich blood away

What Happens During an Asthma Attack

Change	Effect
Bronchioles constrict	Less air reaches the alveoli
Airway lining swells	Further narrows the airway
Mucus production increases	Blocks airflow
Gas exchange decreases	Less oxygen enters the blood; less carbon dioxide leaves
Heart works harder	Must pump faster to deliver available oxygen to tissues

How This Affects the Whole Organism

Level	Effect
Respiratory	Wheezing, coughing, shortness of breath
Circulatory	Increased heart rate; possible decrease in blood oxygen
Nervous system	Anxiety and panic can worsen breathing
Muscular system	Accessory muscles in the neck and chest work harder
Daily life	Difficulty exercising, sleeping, or performing normal activities

Key insight: A problem in the respiratory system directly affects the circulatory system because the two systems work together to deliver oxygen. When breathing is impaired, the heart must compensate, and the entire body suffers from reduced oxygen.

Comparison of the Two Cases

Feature	Torn ACL	Asthma Attack
Primary system affected	Skeletal and muscular	Respiratory
Secondary system affected	Muscular and nervous	Circulatory and nervous
Immediate effect	Knee instability, pain	Difficulty breathing
Systemic effect	Reduced mobility	Reduced oxygen delivery
Treatment	Surgery, physical therapy	Inhalers, medication, avoidance of triggers
Prevention	Strengthening, proper technique	Avoiding triggers, taking controller medication

Part 4: Nervous and Endocrine Regulation of Body Temperature

Why Body Temperature Must Be Regulated

Body temperature affects enzyme activity, cell membrane function, and metabolism. Most human enzymes work best at approximately 37°C. If temperature rises or falls too far, cellular processes fail. The body must maintain a stable internal temperature despite changes in the external environment.

The Two Control Systems

System	How It Communicates	Speed	Duration
Nervous system	Electrical signals through neurons	Very fast (milliseconds)	Short-lived
Endocrine system	Chemical messengers (hormones) through blood	Slower (seconds to hours)	Longer-lasting

The Hypothalamus: The Control Center

The **hypothalamus** is a region of the brain that acts as the body's thermostat. It detects temperature changes and triggers responses through both the nervous and endocrine systems.

Response to Cold: Raising Body Temperature

Step	What Happens	System Involved
1	Thermoreceptors in skin detect cold	Nervous
2	Signal sent to hypothalamus	Nervous
3	Hypothalamus triggers shivering	Nervous
4	Muscles contract rapidly, generating heat	Muscular
5	Blood vessels near skin constrict (vasoconstriction)	Circulatory
6	Less heat lost from skin surface	Circulatory
7	Thyroid releases hormones that increase metabolic rate	Endocrine
8	Adrenal glands release epinephrine and norepinephrine	Endocrine
9	Metabolism increases, producing more heat	Endocrine

Response to Heat: Lowering Body Temperature

Step	What Happens	System Involved
1	Thermoreceptors detect heat	Nervous

Step	What Happens	System Involved
2	Signal sent to hypothalamus	Nervous
3	Hypothalamus triggers sweating	Nervous
4	Sweat glands release sweat; evaporation cools skin	Integumentary
5	Blood vessels near skin dilate (vasodilation)	Circulatory
6	More heat released from skin surface	Circulatory
7	Thyroid hormone levels decrease	Endocrine
8	Metabolic rate slows, producing less heat	Endocrine
9	Behavioral responses (seeking shade, drinking water)	Nervous

Negative Feedback Loop

Both responses follow a **negative feedback loop**: the body detects a change, responds to reverse it, and stops responding once normal temperature is restored.

Component	Role
Sensor	Thermoreceptors in skin and hypothalamus
Control center	Hypothalamus

Component	Role
Effectors	Muscles, blood vessels, sweat glands, thyroid, adrenal glands
Response	Shivering, vasoconstriction, sweating, vasodilation, hormone release
Result	Body temperature returns to normal

Example: Fever as a Reset

During infection, the hypothalamus may reset the body's temperature set point higher. This causes chills (feeling cold) as the body tries to reach the new higher temperature. Once the infection is controlled, the set point returns to normal, and sweating helps release excess heat.

Nervous and Endocrine Interaction Summary

Situation	Nervous Response	Endocrine Response
Cold	Shivering, vasoconstriction	Thyroid hormones, epinephrine increase metabolism
Heat	Sweating, vasodilation	Reduced thyroid hormones lower metabolism
Stress	Rapid heart rate, alertness	Cortisol, epinephrine mobilize energy
Fever	Chills, behavioral changes	Hypothalamic set point adjustment

Part 5: Immune and Lymphatic Systems – How Vaccines Work

The Body's Defense Systems

The body has two interconnected systems for defending against disease: the **immune system** and the **lymphatic system**.

System	Key Components	Function
Immune system	White blood cells (lymphocytes), antibodies, cytokines	Recognize and destroy pathogens
Lymphatic system	Lymph nodes, lymph vessels, spleen, thymus, tonsils	Produce and transport lymphocytes; filter pathogens and waste

How the Immune System Responds to Infection

Line of Defense	Description	Example
First line	Physical and chemical barriers	Skin, mucus, stomach acid
Second line	Innate immune response	Inflammation, fever, phagocytes
Third line	Adaptive immune response	B cells and T cells targeting specific pathogens

Key Cells in Adaptive Immunity

Cell Type	Function
B lymphocytes (B cells)	Produce antibodies that bind to specific antigens
Helper T cells	Activate B cells and cytotoxic T cells
Cytotoxic T cells	Destroy infected cells
Memory B cells	Remain in the body after infection; respond rapidly on re-exposure
Memory T cells	Remain in the body; coordinate rapid response on re-exposure

What Is an Antigen?

An **antigen** is any substance that triggers an immune response. Pathogens such as bacteria and viruses carry antigens on their surfaces.

How Vaccines Work

A **vaccine** contains a harmless form of an antigen — a weakened or inactivated pathogen, a piece of a pathogen, or a toxin that has been modified. The vaccine does not cause disease, but it trains the immune system to recognize the antigen.

Step	What Happens
1. Introduction	Vaccine introduces antigen into the body
2. Recognition	Immune cells detect the antigen as foreign

Step	What Happens
3. Activation	Helper T cells activate B cells and cytotoxic T cells
4. Antibody production	B cells produce antibodies specific to the antigen
5. Memory formation	Memory B cells and memory T cells are produced
6. Resolution	Immune response subsides after antigen is cleared
7. Future exposure	If the real pathogen enters, memory cells respond rapidly and prevent disease

Types of Vaccines

Type	Description	Example
Live attenuated	Weakened form of pathogen	Measles, mumps, rubella (MMR)
Inactivated	Killed pathogen	Polio (IPV), rabies
Subunit	Piece of pathogen (protein)	Hepatitis B, HPV
Toxoid	Inactivated toxin	Tetanus, diphtheria
mRNA	Genetic instructions for antigen	Some COVID-19 vaccines

Type	Description	Example
Viral vector	Harmless virus carries antigen gene	Some COVID-19 vaccines

The Role of the Lymphatic System

Structure	Role in Immunity
Lymph nodes	Filter lymph; contain lymphocytes that detect antigens
Spleen	Filters blood; stores lymphocytes
Thymus	Matures T cells
Tonsils	Trap pathogens entering through mouth and nose
Bone marrow	Produces all blood cells, including lymphocytes

How Vaccines Protect the Community

Concept	Explanation
Herd immunity	When enough people are vaccinated, the pathogen cannot spread easily
Protection of vulnerable	People who cannot be vaccinated (infants, immunocompromised) are protected by those who are

Concept	Explanation
Reduction of outbreaks	Fewer susceptible individuals means fewer opportunities for disease spread

Common Vaccine Myths and Facts

Myth	Fact
Vaccines cause the disease they prevent	Vaccines contain weakened or inactivated pathogens that cannot cause disease in healthy individuals
Vaccines are unnecessary because diseases are rare	Diseases are rare because vaccines work; stopping vaccination allows them to return
Vaccines cause autism	Extensive research has found no link between vaccines and autism
Natural infection is always better	Natural infection carries risks of serious complications and death; vaccines provide immunity without disease

Key Terms

Term	Definition
Xylem	Vascular tissue that transports water and minerals upward in plants
Phloem	Vascular tissue that transports sugars throughout plants
Transpiration	Loss of water vapor through stomata; drives water movement upward
Translocation	Movement of sugars through phloem from source to sink
Cohesion-tension theory	Explanation for water movement in xylem based on evaporation pull
ACL	Anterior cruciate ligament; stabilizes the knee joint
Asthma	Chronic inflammatory disease of the airways
Vasoconstriction	Narrowing of blood vessels; reduces heat loss
Vasodilation	Widening of blood vessels; increases heat loss
Hypothalamus	Brain region that regulates body temperature
Negative feedback	A response that reverses a change to restore balance

Term	Definition
Vaccine	A preparation that stimulates immunity without causing disease
Antigen	A substance that triggers an immune response
Antibody	A protein produced by B cells that binds to a specific antigen
Memory cells	Lymphocytes that remain after infection and provide rapid response on re-exposure
Lymphatic system	Network of vessels and organs that support immune function
Herd immunity	Community protection when a high proportion of individuals are immune

Review Questions

1. Explain how the roots, stem, and leaves of a plant work together to transport water and nutrients.
2. Differentiate xylem and phloem in terms of what they transport and the direction of movement.
3. List five major human body systems and state one function of each.
4. Describe how a torn ACL affects the function of the knee joint and the muscular system.
5. Explain how an asthma attack affects both the respiratory and circulatory systems.
6. Compare the roles of the nervous and endocrine systems in regulating body temperature.
7. Describe the body's response to cold using the negative feedback loop. Identify the sensor, control center, and effectors.
8. Explain how a vaccine trains the immune system to protect against a future infection.
9. Study the scenario below and answer the questions that follow.

A student is diagnosed with a torn ACL after a basketball injury. The doctor recommends surgery and physical therapy. After surgery, the student wears a brace and undergoes months of rehabilitation to strengthen the quadriceps and hamstrings.

- (a) Why is the knee unstable after an ACL tear?
- (b) Why is strengthening the quadriceps and hamstrings part of rehabilitation?
- (c) How does this case demonstrate that a damaged part in one system affects other systems?

10. Application question: A barangay health worker is explaining the importance of vaccination to a group of parents. Some parents are hesitant, saying that vaccines are unnecessary because the diseases they prevent are rare.

- (a) Explain how vaccines work using the functions of the immune and lymphatic systems.
- (b) Explain why diseases like measles and polio are rare today.
- (c) Explain how vaccination protects not only the individual but also the community.
- (d) What would likely happen if vaccination rates dropped significantly?